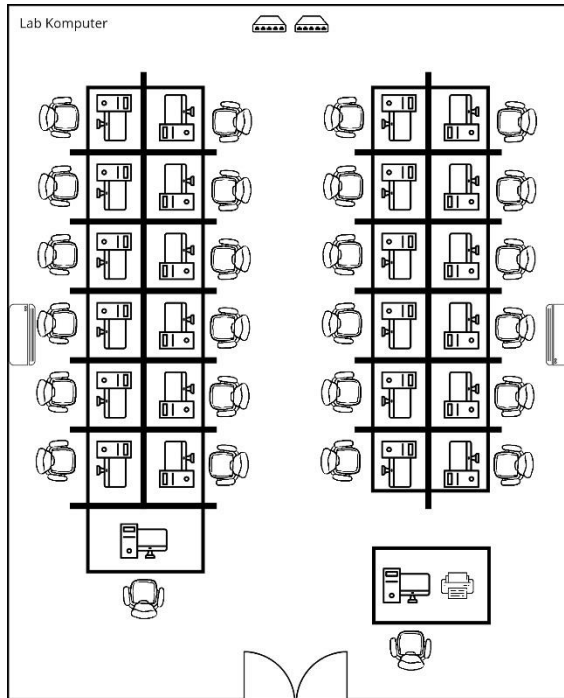


# Dokumentasi Jaringan Sekolah Menengah “Nusantara Jaya”

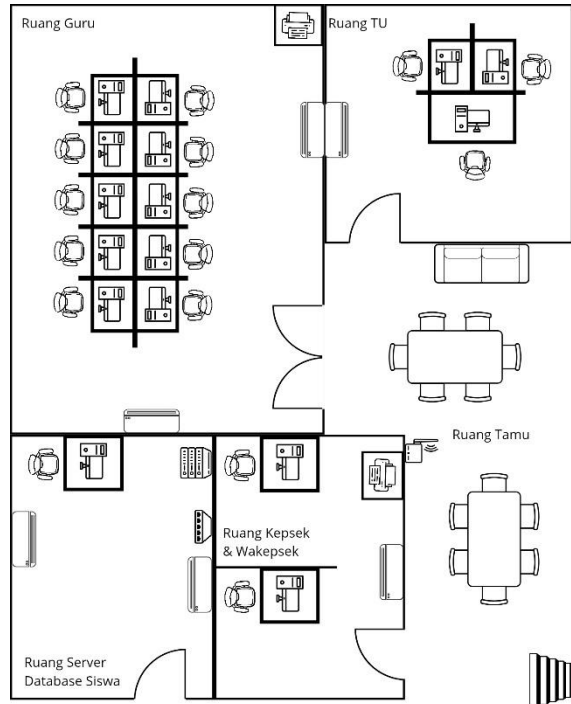
## DAFTAR ISI

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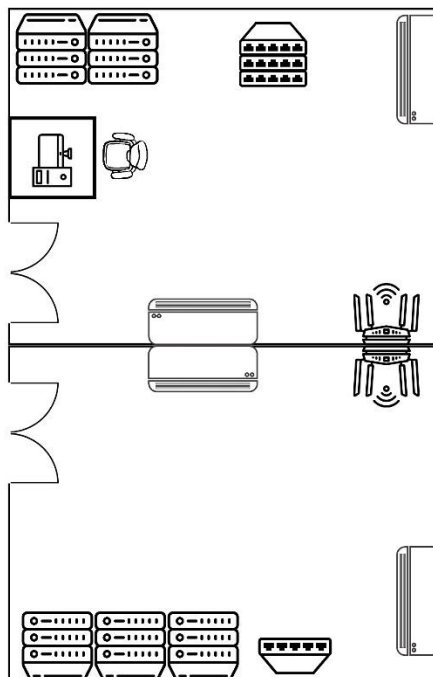
## 1. Denah



**Gedung Utama (A)**  
Lantai 1



**Gedung Utama (A)**  
Lantai 2



**Gedung ICT (B)**  
Lantai Dasar

Denah ini terdiri dari 2 gedung :

- a. Gedung Utama (A) : Gedung A berfungsi sebagai pusat aktivitas pengguna (End-User) yang terdiri dari dua lantai dengan fungsi spesifik. Lantai 1 didedikasikan sebagai Laboratorium Komputer Siswa yang menampung 25 + 1 unit PC dan 1 unit printer. Konektivitas di lantai ini dikelola oleh dua unit Switch Lab yang membagi beban lalu lintas data siswa. Sementara itu, lantai 2 merupakan area manajemen operasional sekolah yang terdiri dari Ruang Guru, Ruang TU, Ruang Kepala dan Wakil Kepala Sekolah, juga Ruang Server Database Siswa. Di area ini terdapat total 15 + 1 unit PC (10 untuk guru, 3 untuk TU, 2 untuk Kepsek dan Wakepsek, dan 1 untuk satpam server), 2 unit printer (printer guru dan printer Kepsek & Wakepsek), serta 1 unit server database siswa lokal untuk keperluan administrasi nilai. Seluruh lalu lintas data dari Switch di Lantai 1 dan Lantai 2 ini diteruskan (uplink) menuju pusat kontrol di Gedung B.
- b. Gedung ICT (B) : Gedung B berperan sebagai "otak" dari keseluruhan topologi jaringan, berfungsi sebagai Data Center dan gerbang internet. Infrastruktur di gedung ini dibagi menjadi dua zona keamanan. Pertama adalah Zona Internal (Intranet), yang berisi Core Switch sebagai titik pertemuan seluruh kabel dari Gedung A, serta Router Gateway Utama. Di zona ini ditempatkan Server E-Learning untuk akses materi pelajaran lokal dan Server Proxy internal untuk manajemen konten. Kedua adalah Zona Simulasi Eksternal (Internet), sebuah ruang terpisah yang mensimulasikan koneksi dunia luar (WAN). Zona ini dilengkapi dengan Router ISP sendiri yang terhubung ke Switch Internet dan menampung server-server publik seperti simulasi Facebook, Instagram, dan TikTok. Pembagian dua zona di Gedung B ini memungkinkan simulasi routing antara jaringan lokal sekolah yang aman dengan jaringan publik internet secara nyata.

## 2. Data Inventaris Ruangan

Gedung A : Lantai 1

Ruang : Laboratorium Komputer

| No | Product Description | Manufacturer | Quantity |
|----|---------------------|--------------|----------|
| 1  | Komputer            | Asus         | 26       |
| 2  | Printer             | Epson        | 1        |
| 3  | Switch              | Cisco        | 2        |

Gedung A : Lantai 2

Ruang : Guru

| No | Product Description | Manufacturer | Quantity |
|----|---------------------|--------------|----------|
| 1  | Komputer            | Asus         | 10       |
| 2  | Printer             | Epson        | 1        |

Gedung A : Lantai 2

Ruang : Tamu

| No | Product Description | Manufacturer | Quantity |
|----|---------------------|--------------|----------|
| 1  | Access Point        | MikroTik     | 1        |

Gedung A : Lantai 2

Ruang : TU (Tata Usaha)

| No | Product Description | Manufacturer | Quantity |
|----|---------------------|--------------|----------|
| 1  | Komputer            | Asus         | 3        |

Gedung A : Lantai 2

Ruang : Kepala Sekolah & Wakil Kepala Sekolah

| No | Product Description | Manufacturer | Quantity |
|----|---------------------|--------------|----------|
| 1  | Komputer            | Asus         | 2        |
| 2  | Printer             | Epson        | 1        |

Gedung A : Lantai 2

Ruang : Server Database Siswa

| No | Product Description | Manufacturer | Quantity |
|----|---------------------|--------------|----------|
| 1  | Komputer            | Asus         | 1        |
| 2  | Switch              | Cisco        | 1        |
| 3  | Server              | Dell         | 1        |

Gedung B : Lantai Dasar

Ruang : Server Intranet

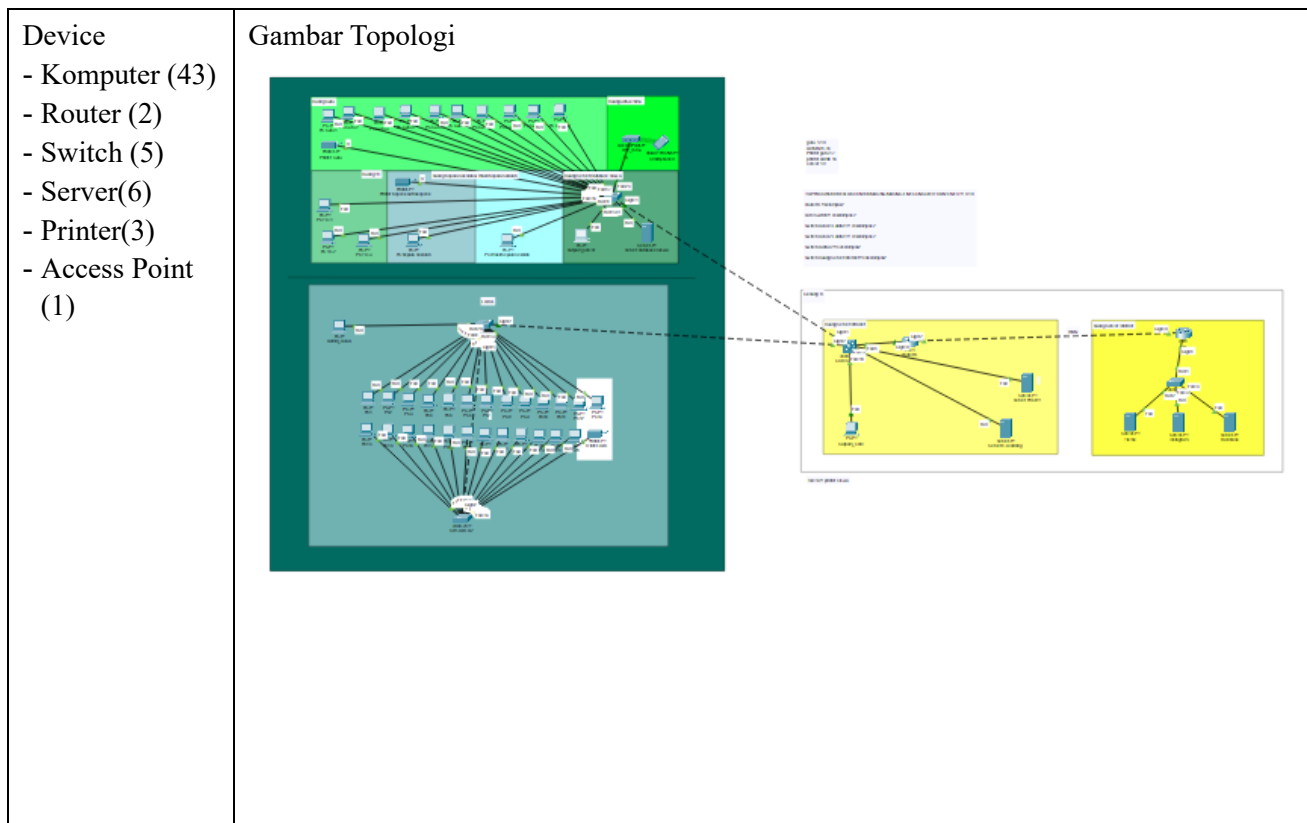
| No | Product Description | Manufacturer | Quantity |
|----|---------------------|--------------|----------|
| 1  | Server              | Dell         | 2        |
| 2  | Router              | MikroTik     | 1        |
| 3  | Core Switch         | Cisco        | 1        |
| 4  | Komputer            | Asus         | 1        |

Gedung B : Lantai Dasar

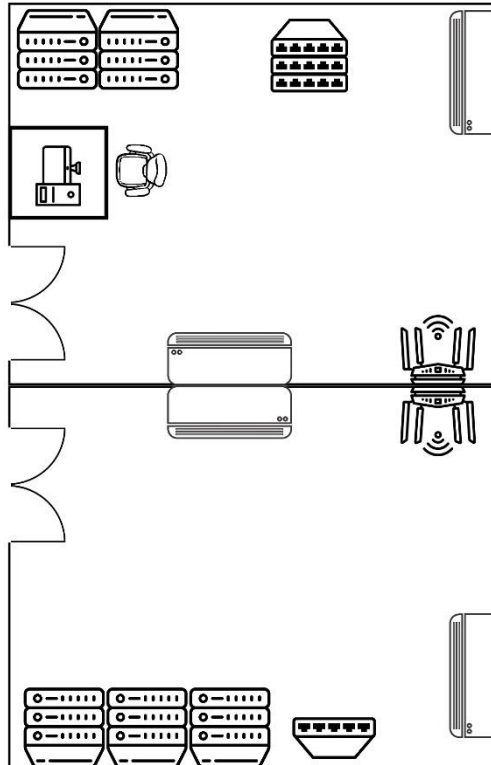
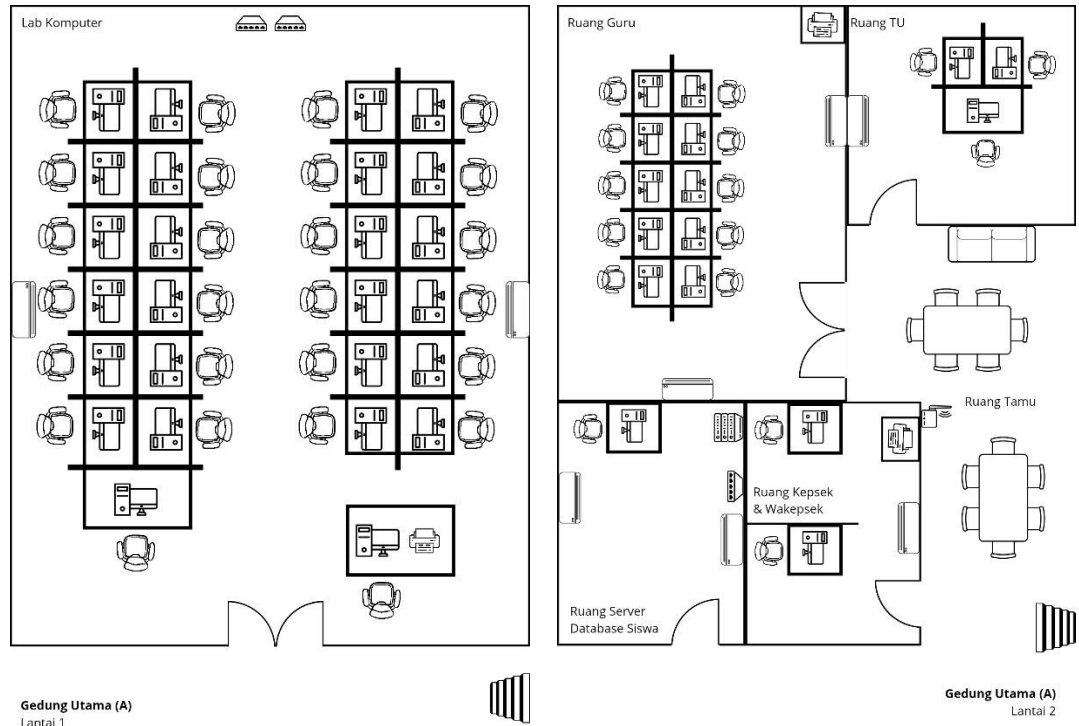
Ruang : Server Internet

| No | Product Description | Manufacturer | Quantity |
|----|---------------------|--------------|----------|
| 1  | Server              | Dell         | 3        |
| 2  | Switch              | Cisco        | 1        |
| 3  | Router              | MikroTik     | 1        |

### 3. Diagram dan Topologi Jaringan Fisik



# Denah



#### 4. Informasi Perangkat Jaringan

| No | Asset Number   | Building         | Floor    | Room         | Manufacturer | IP Address    | Subnet Mask   | Mac Address    |
|----|----------------|------------------|----------|--------------|--------------|---------------|---------------|----------------|
| 1  | PC Admin Lab A | Gedung Utama (A) | Lantai 1 | Lab Komputer | HP           | 192.168.30.11 | 255.255.255.0 | 0030.F28B.7914 |
| 2  | PC 1           |                  |          |              |              | 192.168.10.2  |               | 0030.F28B.7901 |
| 3  | PC 2           |                  |          |              |              | 192.168.10.3  |               | 0030.F28B.7902 |
| 4  | PC 3           |                  |          |              |              | 192.168.10.4  |               | 0030.F28B.7903 |
| 5  | PC 4           |                  |          |              |              | 192.168.10.5  |               | 0030.F28B.7904 |
| 6  | PC 5           |                  |          |              |              | 192.168.10.6  |               | 0030.F28B.7905 |
| 7  | PC 6           |                  |          |              |              | 192.168.10.7  |               | 0030.F28B.7906 |
| 8  | PC 7           |                  |          |              |              | 192.168.10.8  |               | 0030.F28B.7907 |
| 9  | PC 8           |                  |          |              |              | 192.168.10.9  |               | 0030.F28B.7908 |
| 10 | PC 9           |                  |          |              |              | 192.168.10.10 |               | 0030.F28B.7909 |
| 11 | PC 10          |                  |          |              |              | 192.168.10.11 |               | 0030.F28B.790A |
| 12 | PC 11          |                  |          |              |              | 192.168.10.12 |               | 0030.F28B.790B |
| 13 | PC 12          |                  |          |              |              | 192.168.10.13 |               | 0030.F28B.790C |
| 14 | PC 13          |                  |          |              |              | 192.168.10.14 |               | 0030.F28B.790D |
| 15 | PC 14          |                  |          |              |              | 192.168.10.15 |               | 0030.A3C0.DA01 |
| 16 | PC 15          |                  |          |              |              | 192.168.10.16 |               | 0030.A3C0.DA02 |
| 17 | PC 16          |                  |          |              |              | 192.168.10.17 |               | 0030.A3C0.DA03 |
| 18 | PC 17          |                  |          |              |              | 192.168.10.18 |               | 0030.A3C0.DA04 |
| 19 | PC 18          |                  |          |              |              | 192.168.10.19 |               | 0030.A3C0.DA05 |
| 20 | PC 19          |                  |          |              |              | 192.168.10.20 |               | 0030.A3C0.DA06 |
| 21 | PC 20          |                  |          |              |              | 192.168.10.21 |               | 0030.A3C0.DA07 |
| 22 | PC 21          |                  |          |              |              | 192.168.10.22 |               | 0030.A3C0.DA08 |
| 23 | PC 22          |                  |          |              |              | 192.168.10.23 |               | 0030.A3C0.DA09 |
| 24 | PC 23          |                  |          |              |              | 192.168.10.24 |               | 0030.A3C0.DA0A |
| 25 | PC 24          |                  |          |              |              | 192.168.10.25 |               | 0030.A3C0.DA0B |



|    |               |                     |          |              |       |               |               |                   |
|----|---------------|---------------------|----------|--------------|-------|---------------|---------------|-------------------|
| 26 | PC 25         | Gedung Utama<br>(A) | Lantai 1 | Lab Komputer | HP    | 192.168.10.26 | 255.255.255.0 | 0030.A3C0.DA0C    |
| 27 | Printer Lab   |                     |          |              | Epson | -             | -             | 0030.F203.4BAC    |
| 28 | Switch LAB A1 |                     |          |              | Cisco | -             | -             | 00:07:EC:BB:06:5E |
| 29 | Switch LAB A2 |                     |          |              |       | -             | -             | 00:05:5E:C0:EC:93 |

| No | Asset Number | Building            | Floor    | Room     | Manufacturer | IP Address    | Subnet Mask   | Mac Address    |
|----|--------------|---------------------|----------|----------|--------------|---------------|---------------|----------------|
| 1  | PC TU 1      | Gedung Utama<br>(A) | Lantai 2 | Ruang TU | HP           | 192.168.30.12 | 255.255.255.0 | 0001.4399.8B0B |
| 2  | PC TU 2      |                     |          |          |              | 192.168.30.13 |               | 0001.4399.8B0C |
| 3  | PC TU 3      |                     |          |          |              | 192.168.30.14 |               | 0001.4399.8B0D |

| No | Asset Number             | Building            | Floor    | Room                   | Manufacturer | IP Address    | Subnet Mask   | Mac Address    |
|----|--------------------------|---------------------|----------|------------------------|--------------|---------------|---------------|----------------|
| 1  | PC Kepala Sekolah        | Gedung Utama<br>(A) | Lantai 2 | Ruang Kepsek & Wakepek | HP           | 192.168.30.15 | 255.255.255.0 | 0001.4399.8B0E |
| 2  | PC Wakil Kepala Sekolah  |                     |          |                        |              | 192.168.30.16 |               | 0001.4399.8B0F |
| 3  | Printer Kepsek & Wakepek |                     |          |                        | Epson        | -             | -             | 0003.E456.7A63 |

| No | Asset Number          | Building            | Floor    | Room                        | Manufacturer | IP Address    | Subnet Mask   | Mac Address       |
|----|-----------------------|---------------------|----------|-----------------------------|--------------|---------------|---------------|-------------------|
| 1  | PC Satpam Server      | Gedung Utama<br>(A) | Lantai 2 | Ruang Server Database Siswa | HP           | 192.168.30.17 | 255.255.255.0 | 0001.4399.8B10    |
| 2  | Switch                |                     |          |                             | Cisco        | -             | -             | 00:60:70:A4:D4:89 |
| 3  | Server Database Siswa |                     |          |                             | Dell         | -             | -             | 000A.F3BE.7C92    |

| No | Asset Number | Building         | Floor    | Room       | Manufacturer | IP Address    | Subnet Mask   | Mac Address    |
|----|--------------|------------------|----------|------------|--------------|---------------|---------------|----------------|
| 1  | PC Guru 1    | Gedung Utama (A) | Lantai 2 | Ruang Guru | HP           | 192.168.20.2  | 255.255.255.0 | 0001.4399.8B01 |
| 2  | PC Guru 2    |                  |          |            |              | 192.168.20.3  |               | 0001.4399.8B02 |
| 3  | PC Guru 3    |                  |          |            |              | 192.168.20.4  |               | 0001.4399.8B03 |
| 4  | PC Guru 4    |                  |          |            |              | 192.168.20.5  |               | 0001.4399.8B04 |
| 5  | PC Guru 5    |                  |          |            |              | 192.168.20.6  |               | 0001.4399.8B05 |
| 6  | PC Guru 6    |                  |          |            |              | 192.168.20.7  |               | 0001.4399.8B06 |
| 7  | PC Guru 7    |                  |          |            |              | 192.168.20.8  |               | 0001.4399.8B07 |
| 8  | PC Guru 8    |                  |          |            |              | 192.168.20.9  |               | 0001.4399.8B08 |
| 9  | PC Guru 9    |                  |          |            |              | 192.168.20.10 |               | 0001.4399.8B09 |
| 10 | PC Guru 10   |                  |          |            |              | 192.168.20.11 |               | 0001.4399.8B0A |
| 11 | Printer Guru |                  |          |            | Epson        | -             | -             | 00D0.9727.3A3C |

| No | Asset Number           | Building         | Floor    | Room             | Manufacturer | IP Address | Subnet Mask | Mac Address |
|----|------------------------|------------------|----------|------------------|--------------|------------|-------------|-------------|
| 1  | Access Point WIFI Tamu | Gedung Utama (A) | Lantai 2 | Ruang untuk Tamu | MikroTik     | -          | -           | -           |

| No | Asset Number          | Building       | Floor    | Room                  | Manufacturer | IP Address    | Subnet Mask   | Mac Address    |
|----|-----------------------|----------------|----------|-----------------------|--------------|---------------|---------------|----------------|
| 1  | Core Switch           | Gedung ICT (B) | Lantai 1 | Ruang Server Intranet | Cisco        | -             | -             | 0004.9ACD.4459 |
| 2  | PC Satpam Core Switch |                |          |                       | HP           | 192.168.30.18 | 255.255.255.0 | 0030.A33D.E6BE |
| 3  | Server Proxy          |                |          |                       | Dell         | -             | -             | 0060.4737.E63D |
| 4  | Server E-Learning     |                |          |                       |              | -             | -             | 0000.0C2D.0B8E |
| 5  | Router B              |                |          |                       | MikroTik     | -             | -             | 0060.7009.A401 |

| No | Asset Number     | Building       | Floor    | Room                  | Manufacturer | IP Address | Subnet Mask | Mac Address       |
|----|------------------|----------------|----------|-----------------------|--------------|------------|-------------|-------------------|
| 1  | Router (ISP)     | Gedung ICT (B) | Lantai 1 | Ruang Server Internet | MikroTik     | -          | -           | 0000.0C94.C501    |
| 2  | Switch           |                |          |                       | Cisco        | -          | -           | 00:30:A3:C7:13:BA |
| 3  | Server Tiktok    |                |          |                       | Dell         | -          | -           | 0001.C9D2.5E28    |
| 4  | Server Instagram |                |          |                       |              | -          | -           | 0000.0C83.EB75    |
| 5  | Server Facebook  |                |          |                       |              | -          | -           | 0001.97E9.0B4E    |

## 5. Labelling

Asset Number : SW-LAB A1

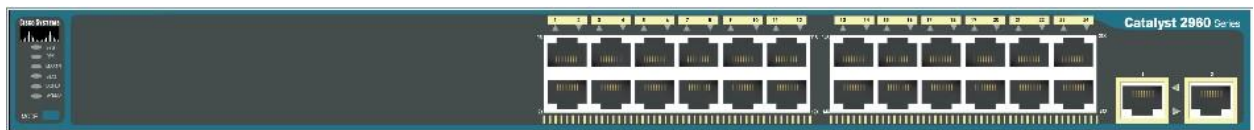
Serial Number : FOC1010X104

Room : Lab A

Product Description : Cisco 2960

Line console 0 : s1kelompok2

Line vty 0 15 : s1kelompok2



| No | Port    | Perangkat Tujuan | IP Address Perangkat | IP Address Tujuan | Keterangan     |
|----|---------|------------------|----------------------|-------------------|----------------|
| 1  | Fa 0/1  | PC Lab 1         | 192.168.10.2         | 192.168.10.1      | Access VLAN 10 |
| 2  | Fa 0/2  | PC Lab 2         | 192.168.10.3         | 192.168.10.1      | Access VLAN 10 |
| 3  | Fa 0/3  | PC Lab 3         | 192.168.10.4         | 192.168.10.1      | Access VLAN 10 |
| 4  | Fa 0/4  | PC Lab 4         | 192.168.10.5         | 192.168.10.1      | Access VLAN 10 |
| 5  | Fa 0/5  | PC Lab 5         | 192.168.10.6         | 192.168.10.1      | Access VLAN 10 |
| 6  | Fa 0/6  | PC Lab 6         | 192.168.10.7         | 192.168.10.1      | Access VLAN 10 |
| 7  | Fa 0/7  | PC Lab 7         | 192.168.10.8         | 192.168.10.1      | Access VLAN 10 |
| 8  | Fa 0/8  | PC Lab 8         | 192.168.10.9         | 192.168.10.1      | Access VLAN 10 |
| 9  | Fa 0/9  | PC Lab 9         | 192.168.10.10        | 192.168.10.1      | Access VLAN 10 |
| 10 | Fa 0/10 | PC Lab 10        | 192.168.10.11        | 192.168.10.1      | Access VLAN 10 |
| 11 | Fa 0/11 | PC Lab 11        | 192.168.10.12        | 192.168.10.1      | Access VLAN 10 |

|    |         |              |               |              |                |
|----|---------|--------------|---------------|--------------|----------------|
| 12 | Fa 0/12 | PC Lab 12    | 192.168.10.13 | 192.168.10.1 | Access VLAN 10 |
| 13 | Fa 0/13 | PC Lab 13    | 192.168.10.14 | 192.168.10.1 | Access VLAN 10 |
| 14 | Fa 0/20 | PC Admin Lab | 192.168.30.11 | 192.168.30.1 | Access VLAN 30 |
| 15 | Gig 0/1 | SW-LAB A2    | -             | -            | Trunk          |

Asset Number : SW-LAB A2

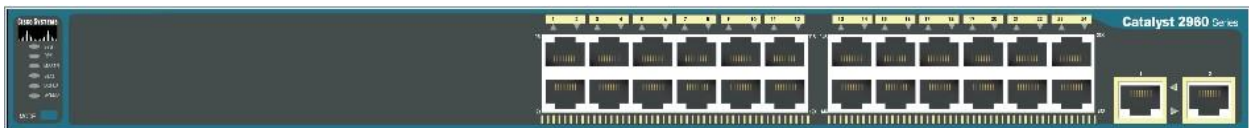
Serial Number : FOC1010X104

Room : Lab A

Product Description : Cisco 2960

Line console 0 : s1kelompok2

Line vty 0 15 : s1kelompok2



| No | Port   | Perangkat Tujuan | IP Address Perangkat | IP Address Tujuan | Keterangan     |
|----|--------|------------------|----------------------|-------------------|----------------|
| 1  | Fa 0/1 | PC Lab 14        | 192.168.10.15        | 192.168.10.1      | Access VLAN 10 |
| 2  | Fa 0/2 | PC Lab 15        | 192.168.10.16        | 192.168.10.1      | Access VLAN 10 |
| 3  | Fa 0/3 | PC Lab 16        | 192.168.10.17        | 192.168.10.1      | Access VLAN 10 |
| 4  | Fa 0/4 | PC Lab 17        | 192.168.10.18        | 192.168.10.1      | Access VLAN 10 |
| 5  | Fa 0/5 | PC Lab 18        | 192.168.10.19        | 192.168.10.1      | Access VLAN 10 |
| 6  | Fa 0/6 | PC Lab 19        | 192.168.10.20        | 192.168.10.1      | Access VLAN 10 |
| 7  | Fa 0/7 | PC Lab 20        | 192.168.10.21        | 192.168.10.1      | Access VLAN 10 |

|    |         |             |               |              |                |
|----|---------|-------------|---------------|--------------|----------------|
| 8  | Fa 0/8  | PC Lab 21   | 192.168.10.22 | 192.168.10.1 | Access VLAN 10 |
| 9  | Fa 0/9  | PC Lab 22   | 192.168.10.23 | 192.168.10.1 | Access VLAN 10 |
| 10 | Fa 0/10 | PC Lab 23   | 192.168.10.24 | 192.168.10.1 | Access VLAN 10 |
| 11 | Fa 0/11 | PC Lab 24   | 192.168.10.25 | 192.168.10.1 | Access VLAN 10 |
| 12 | Fa 0/12 | PC Lab 25   | 192.168.10.26 | 192.168.10.1 | Access VLAN 10 |
| 13 | Fa 0/13 | Printer Lab | -             | -            | Access VLAN 10 |

Asset Number : SW-2

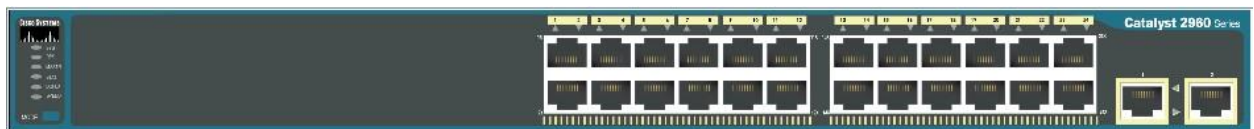
Serial Number : FOC1010X104

Room : Server Database Siswa

Product Description : Cisco 2960

Line console 0 : s2kelompok2

Line vty 0 15 : s2kelompok2



| No | Port   | Perangkat Tujuan | IP Address Perangkat | IP Address Tujuan | Keterangan     |
|----|--------|------------------|----------------------|-------------------|----------------|
| 1  | Fa 0/1 | PC Guru 1        | 192.168.20.2         | 192.168.20.1      | Access VLAN 20 |
| 2  | Fa 0/2 | PC Guru 2        | 192.168.20.3         | 192.168.20.1      | Access VLAN 20 |
| 3  | Fa0/3  | PC Guru 3        | 192.168.20.4         | 192.168.20.1      | Access VLAN 20 |
| 4  | Fa0/5  | PC Guru 4        | 192.168.20.5         | 192.168.20.1      | Access VLAN 20 |

|    |        |                              |                |              |                   |
|----|--------|------------------------------|----------------|--------------|-------------------|
| 5  | Fa0/4  | PC Guru 5                    | 192.168.20.6   | 192.168.20.1 | Access VLAN<br>20 |
| 6  | Fa0/6  | PC Guru 6                    | 192.168.20.7   | 192.168.20.1 | Access VLAN<br>20 |
| 7  | Fa0/7  | PC Guru 7                    | 192.168.20.8   | 192.168.20.1 | Access VLAN<br>20 |
| 8  | Fa0/8  | PC Guru 8                    | 192.168.20.9   | 192.168.20.1 | Access VLAN<br>20 |
| 9  | Fa0/9  | PC Guru 9                    | 192.168.20.10  | 192.168.20.1 | Access VLAN<br>20 |
| 10 | Fa0/10 | PC Guru 10                   | 192.168.20.11  | 192.168.20.1 | Access VLAN<br>20 |
| 11 | Fa0/11 | PC TU 1                      | 192.168.30. 12 | 192.168.30.1 | Access VLAN<br>30 |
| 12 | Fa0/12 | PC TU 2                      | 192.168.30. 13 | 192.168.30.1 | Access VLAN<br>30 |
| 13 | Fa0/13 | PC TU 3                      | 192.168.30. 14 | 192.168.30.1 | Access VLAN<br>30 |
| 14 | Fa0/14 | PC Kepala Sekolah            | 192.168.30.15  | 192.168.30.1 | Access VLAN<br>30 |
| 15 | Fa0/15 | PC Wakil Kepala<br>Sekolah   | 192.168.30.16  | 192.168.30.1 | Access VLAN<br>30 |
| 16 | Fa0/16 | Printer Kepsek &<br>Wakepsek | -              | -            | Access VLAN<br>30 |
| 17 | Fa0/17 | Printer Guru                 | -              | -            | Access VLAN<br>20 |
| 18 | Fa0/18 | Server Database<br>Siswa     | -              | -            | Access VLAN<br>30 |
| 19 | Fa0/19 | PC Satpam Server             | 192.168.30.17  | 192.168.30.1 | Access VLAN<br>30 |
| 21 | Fa0/24 | Access Point Ruang<br>Tamu   | -              | -            | Access VLAN<br>99 |
| 22 | Gig0/1 | Core Switch                  | -              | -            | Trunk             |

Asset Number : Core Switch B

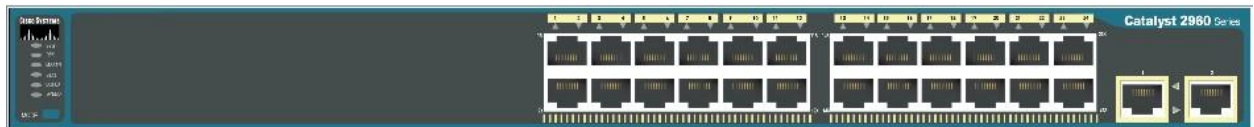
Serial Number : CAT1037RJF7

Room : Server Intranet

Product Description : 3560 24PS

Line console 0 : cskelompok2

Line vty 0 15 : cskelompok2



| No | Port   | Perangkat Tujuan | IP Address Perangkat | IP Address Tujuan | Keterangan     |
|----|--------|------------------|----------------------|-------------------|----------------|
| 1  | Fa 0/1 | PC Guru 1        | 192.168.20.2         | 192.168.20.1      | Access VLAN 20 |
| 2  | Fa 0/2 | PC Guru 2        | 192.168.20.3         | 192.168.20.1      | Access VLAN 20 |



Konfigurasi (berlaku untuk semua device switch dan router, hanya saja menyesuaikan line console 0 dan line vty nya):

```
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#enable secret cskelompok2
Switch(config)#line console 0
Switch(config-line)#password cskelompok2
Switch(config-line)#login
Switch(config-line)#exit
Switch(config)#line vty 0 15
Switch(config-line)#password cskelompok2
Switch(config-line)#login
Switch(config-line)#exit
Switch(config)#service password-encryption
Switch(config)#do write
Building configuration...
[OK]
Switch(config)#
```

#### User Access Verification

Password:

```
Switch>enable
Password:
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name SISWA
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name GURU
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name ADMIN
Switch(config-vlan)#exit
Switch(config)#vlan 40
Switch(config-vlan)#name SERVER
Switch(config-vlan)#exit
Switch(config)#interface range gigabitEthernet 0/1-2, fastEthernet 0/1
Switch(config-if-range)#switchport trunk encapsulation dot1q
Switch(config-if-range)#switchport mode trunk

Switch(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if-range)#no shutdown
Switch(config-if-range)#exit
Switch(config)#
```

```
Switch(config)#interface range fastEthernet 0/2-5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 40
Switch(config-if-range)#no shutdown
Switch(config-if-range)#exit
Switch(config)#do write
Building configuration...
[OK]
Switch(config)#
```

Switch>

Switch con0 is now available

Press RETURN to get started.

```
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name SISWA
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name GURU
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name ADMIN
Switch(config-vlan)#exit
Switch(config)#vlan 40
Switch(config-vlan)#name SERVER
Switch(config-vlan)#exit
Switch(config)#interface range fastEthernet 0/1-10, fastEthernet 0/17
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#no shutdown
Switch(config-if-range)#exit
Switch(config)#interface range fastEthernet 0/11-15, fastEthernet 0/16, fastEthernet 0/18
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#no shutdown
Switch(config-if-range)#exit
Switch(config)#interface gigabitEthernet 0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#do write
Building configuration...
[OK]
Switch(config)#
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
```

```
Switch>
```

```
|
```

```
Switch con0 is now available
```

```
Press RETURN to get started.
```

```
Switch>enable
```

```
Switch#conf t
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
Switch(config)#vlan 10
```

```
Switch(config-vlan)#name SISWA
```

```
Switch(config-vlan)#exit
```

```
Switch(config)#vlan 20
```

```
Switch(config-vlan)#name GURU
```

```
Switch(config-vlan)#exit
```

```
Switch(config)#vlan 30
```

```
Switch(config-vlan)#name ADMIN
```

```
Switch(config-vlan)#exit
```

```
Switch(config)#vlan 40
```

```
Switch(config-vlan)#name SERVER
```

```
Switch(config-vlan)#exit
```

```
Switch(config)#interface rang3 fastEthernet 0/1-24
```

```
% Invalid input detected at '^' marker.
```

```
Switch(config)#interface range fastEthernet 0/1-24
```

```
Switch(config-if-range)#switchport mode access
```

```
Switch(config-if-range)#switchport access vlan 10
```

```
Switch(config-if-range)#exit
```

```
Switch(config)#interface range gigabitEthernet 0/1-2
```

```
Switch(config-if-range)#switchport mode trunk
```

```
Switch(config-if-range)#exit
```

```
Switch(config)#do write
```

```
Building configuration...
```

```
[OK]
```

```
- . . . . .
```

```

Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/0.10
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.20
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.30
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up

Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.40
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.40, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.40, changed state to up

Router(config-subif)#encapsulation dot1Q 40
Router(config-subif)#ip address 192.168.40.1 255.255.255.0
Router(config-subif)#exit
Router(config)#ip dhcp pool POOL_SISWA
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp pool POOL_GURU
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp pool POOL_ADMIN
Router(dhcp-config)#network 192.168.30.0 255.255.255.0
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#

```

```

Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/0.10
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.20
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.30
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up

Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.40
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.40, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.40, changed state to up

Router(config-subif)#encapsulation dot1Q 40
Router(config-subif)#ip address 192.168.40.1 255.255.255.0
Router(config-subif)#exit
Router(config)#ip dhcp pool POOL_SISWA
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp pool POOL_GURU
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp pool POOL_ADMIN
Router(dhcp-config)#network 192.168.30.0 255.255.255.0
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#

```

```

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.30
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up

Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.40
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.40, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.40, changed state to up

Router(config-subif)#encapsulation dot1Q 40
Router(config-subif)#ip address 192.168.40.1 255.255.255.0
Router(config-subif)#exit
Router(config)#ip dhcp pool POOL_SISWA
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp pool POOL_GURU
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp pool POOL_ADMIN
Router(dhcp-config)#network 192.168.30.0 255.255.255.0
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip access-list extended BATAS_SISWA
Router(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
Router(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
Router(config-ext-nacl)#deny tcp 192.168.10.0 0.0.0.255 ^ 192.168.40.10 eq 80

% Invalid input detected at '^' marker.

Router(config-ext-nacl)#deny tcp 192.168.10.0 0.0.0.255 host 192.168.40.10 eq 80
Router(config-ext-nacl)#permit ip any any
^

% Invalid input detected at '^' marker.

Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#exit
Router(config)#interface gigabitEthernet 0/0.10
Router(config-subif)#ip access-group BATAS_SISWA in
Router(config-subif)#exit
Router(config)#ip access-list extended BATAS_ADMIN
Router(config-ext-nacl)#deny ip 192.168.30.0 0.0.0.255 host 192.168.40.20
Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#exit
Router(config)#interface gigabitEthernet 0/0.30
Router(config-subif)#ip access-group BATAS_ADMIN in
Router(config-subif)#exit
Router(config)#do write
Building configuration...
[OK]
Router(config)#

```

```

Router(config)#ip access-list extended BATAS_SISWA
Router(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
Router(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
Router(config-ext-nacl)#deny tcp 192.168.10.0 0.0.0.255 host 192.168.40.10 eq 80
^
% Invalid input detected at '^' marker.

Router(config-ext-nacl)#deny tcp 192.168.10.0 0.0.0.255 host 192.168.40.10 eq 80
Router(config-ext-nacl)#permit ip any any
^
% Invalid input detected at '^' marker.

Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#exit
Router(config)#interface gigabitEthernet 0/0.10
Router(config-subif)#ip access-group BATAS_SISWA in
Router(config-subif)#exit
Router(config)#ip access-list extended BATAS_ADMIN
Router(config-ext-nacl)#deny ip 192.168.30.0 0.0.0.255 host 192.168.40.20
Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#exit
Router(config)#interface gigabitEthernet 0/0.30
Router(config-subif)#ip access-group BATAS_ADMIN in
Router(config-subif)#exit
Router(config)#do write
Building configuration...
[OK]
Router(config)#

```

---

```

24 FastEthernet interfaces
2 Gigabit Ethernet interfaces
The password-recovery mechanism is enabled.
64K bytes of flash-simulated non-volatile configuration memory.
Base ethernet MAC Address       : 00:30:A3:C7:13:BA
Motherboard assembly number     : 73-10390-03
Power supply part number        : 341-0097-02
Motherboard serial number       : FOC10093R12
Power supply serial number      : AZS1007032H
Model revision number           : B0
Motherboard revision number     : B0
Model number                    : WS-C2960-24TT-L
System serial number            : FOC1010X104
Top Assembly Part Number        : 800-27221-02
Top Assembly Revision Number    : A0
Version ID                      : V02
CLEI Code Number                : COM3L00BRA
Hardware Board Revision Number  : 0x01

Switch Ports Model          SW Version  SW Image
-----
*   1 26   WS-C2960-24TT-L   15.0(2)SE4   C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnnguyen

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

Switch>
Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
% Access VLAN does not exist. Creating vlan 40
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#do write
Building configuration...
[OK]
Switch(config)#

```

```

Switch>
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
% Access VLAN does not exist. Creating vlan 40
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#do write
Building configuration...
[OK]
Switch(config)#

```

Asset Number : Router B

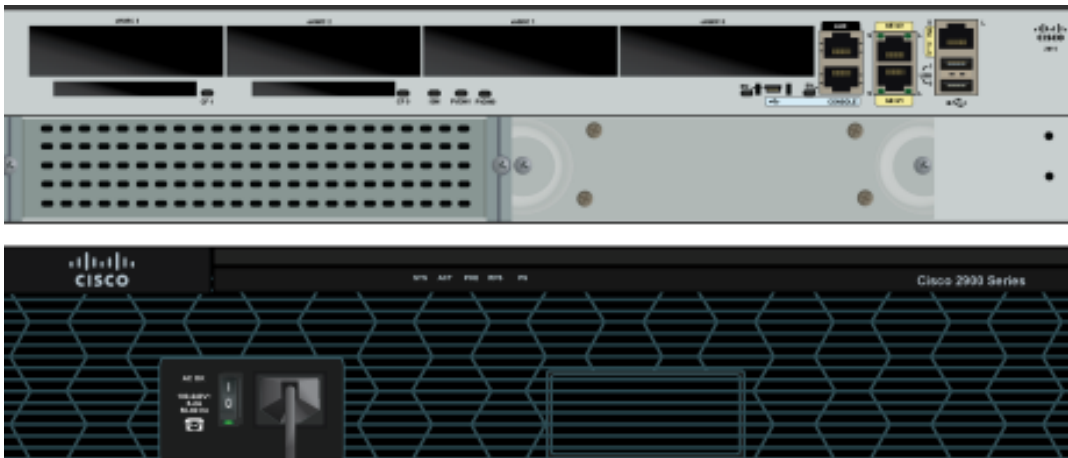
Serial Number : FTX152400KS

Room : Server Intranet

Product Description : Router 2911

Line console 0 : kelompok2

Line vty 0 15 : kelompok2



| No | Port    | Perangkat Tujuan | IP Address Perangkat | IP Address Tujuan | Keterangan                     |
|----|---------|------------------|----------------------|-------------------|--------------------------------|
| 1  | Gig 0/0 | Core Switch      | -                    | -                 | Access VLAN 10, 20, 30, 40, 99 |
| 2  | Gig 0/2 | Router ISP       | -                    | 200.200.200.1     | Routing ke Internet (WAN)      |



Asset Number : Router (ISP)

Serial Number : FTX152400KS

Room : Server Internet

Product Description : Router 2911

Line console 0 : kelompok2

Line vty 0 15 : kelompok2



| No | Port    | Perangkat Tujuan | IP Address Perangkat | IP Address Tujuan | Keterangan               |
|----|---------|------------------|----------------------|-------------------|--------------------------|
| 1  | Gig 0/0 | Router B         | -                    | 200.200.200.2     | Routing ke Sekolah (WAN) |
| 2  | Gig 0/1 | Switch0          | -                    | 8.8.8.1           | Gateway Server Internet  |

Asset Number : Switch0

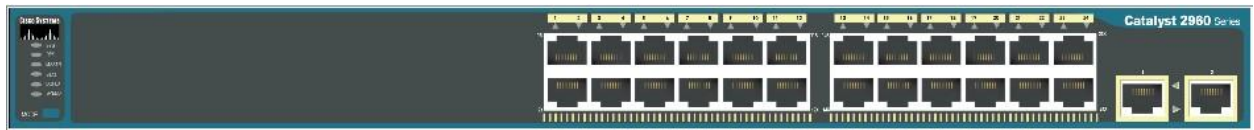
Serial Number : FOC1010X104

Room : Server Internet

Product Description : Cisco 2960

Line console 0 : s0kelompok2

Line vty 0 15 : s0kelompok2



| No | Port   | Perangkat Tujuan | IP Address Perangkat | IP Address Tujuan | Keterangan             |
|----|--------|------------------|----------------------|-------------------|------------------------|
| 1  | Fa 0/1 | Router (ISP)     | -                    | -                 | Access VLAN 1 (Uplink) |
| 2  | Fa 0/2 | Server TikTok    | -                    | -                 | Access VLAN 1          |
| 3  | Fa 0/3 | Server Instagram | -                    | -                 | Access VLAN 1          |
| 4  | Fa 0/4 | Server Facebook  | -                    | -                 | Access VLAN 1          |